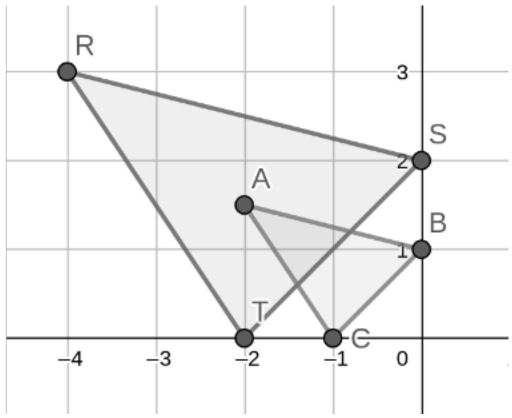
Grade 8 Name____________________
Unit 9 Date____________________
Lesson 9 Practice Period___________________

On the coordinate grid shown, $\triangle RST$ is the result of dilating $\triangle ABC$ using the origin as the center of dilation.



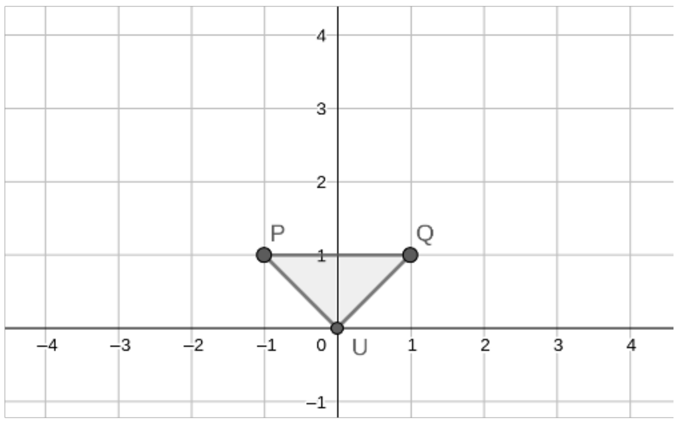
1. Alana states "The scale factor of this dilation is $\frac{1}{2}$. To find the scale factor, I found the coordinates of point R which were $(-4, 3)$ and the coordinates of point A which were $(-2, 1.5)$. I know that $-4 \times \frac{1}{2} = -2$ and $3 \times \frac{1}{2} = 1.5$, so the scale factor is $\frac{1}{2}$." Explain Alana's error.

Using the coordinate grid above, complete the table with the coordinates of each vertex.

Coordinates		Coordinates	
2. Point *A*		**Point *R***	
3. Point *B*		**Point *S***	
4. Point *C*		**Point *T***	

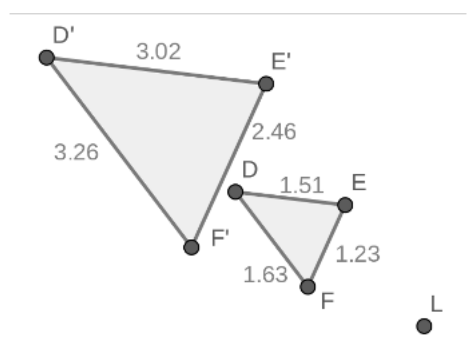
5. What is the scale factor use to dilate $\triangle ABC$?

Archaeologists often use LIDAR (light detection and ranging) to map areas beneath the ground during their excavation of ancient cities. Utilizing coordinate grids to plot these artifacts allows them to analyze the various objects being excavated. An artifact in the shape of $\triangle PQU$ shown on the grid below was found. The LIDAR uses a zoom feature that dilates the artifacts so they are easier to see. Grant knows if he dilates $\triangle PQU$ using a scale factor of 4 centered at the origin, he will have the true coordinates of the artifact. Determine what the coordinates of the artifact $\triangle P'Q'U'$ would be and plot $\triangle P'Q'U'$ on the coordinate grid below.



Coordinates		Coordinates	
6. Point P		**Point P'**	
7. Point Q		**Point Q'**	
8. Point U		**Point U'**	

9. The figure shows the result of dilating $\triangle DEF$ using point L as the center of dilation. Measurements, in units, of each side are given. Determine the scale factor of the dilation.



10. Using the figure in question 9, if $\triangle DEF$ was dilated using a scale factor of 1.2, what would the length of EF measure?